

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Inorganic Cation Transmembrane Transport	0.09203326	277	1.548e-07	8.339e-04	ATP6V0A2:16 KCNK18:30 ATP2A1:47 KCNA4:114 SLC6A6:115 KCNQ5:125
Monosaccharide Cation Transmembrane Transport	0.08776057	273	6.773e-07	1.825e-03	ATP6V0A2:16 KCNK18:30 ATP2A1:47 KCNA4:114 SLC6A6:115 KCNQ5:125
Carboxylic Acid Transport (GO:0046942)	0.16850755	50	3.803e-05	5.122e-02	SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A19:402 SLC6A14:588
Potassium Ion Transmembrane Transport (G	0.10567398	132	2.872e-05	5.122e-02	KCNK18:30 KCNJ16:53 KCNA4:114 KCNQ5:125 LRRCS5:159 KCNJ1:213
Monocarboxylic Acid Transport (GO:001571)	0.14653168	63	5.854e-05	6.158e-02	SLC27A4:37 TSP02:126 ATP8B1:176 SLC10A2:250 MFSD2A:341 ABCD3:41
Nitrogen Compound Transport (GO:0071705)	0.09409977	151	6.857e-05	6.158e-02	SLC43A1:68 SLC6A6:115 SLC22A4:153 RD3:174 SLC6A15:192 SLC22A1:240
NADH Dehydrogenase Complex Assembly (GO:	-0.16308881	48	9.366e-05	6.308e-02	NDUFA11:192 NDUFB10:305 NDUFA8:471 TMEM126B:500 NDUFB8:505 OXA1L:630
Mitochondrial Respiratory Chain Complex	-0.16308881	48	9.366e-05	6.308e-02	NDUFA11:192 NDUFB10:305 NDUFA8:471 TMEM126B:500 NDUFB8:505 OXA1L:630
Axoneme Assembly (GO:0035082)	-0.20440312	30	1.073e-04	6.398e-02	CCDC40:29 TTC12:302 RSPH6A:417 RPL1:477 CFAP74:695 SPAG1:842
Fatty Acid Transport (GO:0015908)	0.18292934	37	1.187e-04	6.398e-02	SLC27A4:37 CD36:187 SLC22A1:240 FABP9:317 MFSD2A:341 FABP1:472
Oxidative Phosphorylation (GO:0006119)	-0.14448858	55	2.125e-04	1.041e-01	ATP5PD:113 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450 ATP5MG:452
Defense Response To Bacterium (GO:0042725)	-0.10003435	113	2.456e-04	1.103e-01	NLRP1:32 DEFB132:150 IL18:355 RNASEH:490 TLR6:498 AZU1:588
Fatty Acid Catabolic Process (GO:0009062)	0.13686037	58	3.157e-04	1.121e-01	SLC27A4:37 EC12:79 LPIN2:137 DECR1:189 ABCD3:374 ACOT8:524
One-Carbon Compound Transport (GO:0019175)	0.19238848	29	3.375e-04	1.121e-01	SLC4A3:112 SLC4A1:146 SLC4A7:323 CA2:392 SLC12A2:408 UPK3A:544
Proton Motive Force-Driven Mitochondrial	-0.15151399	47	3.290e-04	1.121e-01	ATP5PD:113 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450 ATP5MG:452
Cilium-Dependent Cell Motility (GO:00602)	-0.15478225	17	4.035e-04	1.359e-01	DNAH2:36 TEK1:1320 CFAP54:548 DNAH3:654 DNAH1:677 DNAH8:739
Amino Acid Transport (GO:0006865)	0.15246805	43	4.566e-04	1.633e-01	SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A19:402 SLC6A14:588
piRNA Processing (GO:0034587)	0.23572407	18	5.366e-04	1.633e-01	GPA2:49 ANKRD34B:104 TDRKH:389 PNLD1:514 FKBP6:687 DD4:954
Gamma-Aminobutyric Acid Transport (GO:00	0.39932000	6	7.056e-04	2.001e-01	SLC6A6:115 SLC6A1:768 SLC6A12:964 SLC6A11:1806 SLC6A13:2838 SLC9A3R1:3008
Amino Acid Import Across Plasma Membrane	0.19320087	25	8.292e-04	2.185e-01	SLC1A1:64 SLC6A6:115 TSP02:126 SLC22A4:153 SLC6A14:588 SLC38A5:653
Peptide Biosynthetic Process (GO:0043043)	-0.08726391	122	8.923e-04	2.185e-01	MRPL24:154 GGT6:255 MRPL37:388 CARN5:1400 DMD:445 PABPC4:517
Potassium Ion Transport (GO:0006813)	0.08837742	119	8.886e-04	2.185e-01	KCNK18:30 KCNJ16:53 KCNA4:114 KCNQ5:125 LRRCS5:159 KCNH4:214
Fatty Acid Beta-Oxidation (GO:0006635)	0.13785216	47	1.085e-03	2.436e-01	EC12:79 GCDH:127 DECR1:189 ABCD3:374 ACOT8:524 PEX2:797
Regulation Of Type II Interferon Product	-0.11111668	73	1.042e-03	2.436e-01	PTPN22:167 IL18:355 RIPK3:432 CD14:448 CD244:513 TLR3:713
Cilium Movement (GO:0003341)	-0.13684630	47	1.180e-03	2.543e-01	CCDC40:29 DNAI2:305 CFAP2:5160 TEK1:1320 RSPH6A:417 CLFAP5:4548
Mitochondrial Respiratory Chain Complex	-0.10646919	77	1.254e-03	2.599e-01	NDUFA11:192 NDUFB10:305 LYRM7:405 NDUFA8:471 TMEM126B:500 NDUFB8:505
T Cell Mediated Immunity (GO:0002455)	-0.24490531	14	1.512e-03	2.733e-01	C4BPA:488 EMP2:612 PTN3A:734 CDBA:913 CD70:997 MYO1G:1021
Aerobic Respiration (GO:0009060)	-0.12408666	54	1.623e-03	2.733e-01	NDUFA11:192 NDUFB10:305 NDUFV3:360 NDUFA8:471 NDUFB8:505 OXA1L:630
Defense Response To Symbiont (GO:0140546)	-0.09090314	101	1.623e-03	2.733e-01	NLRP1:32 CD207:35 NLRP9:64 CD40:237 MX2:278 RIKP3:43
Long-Chain Fatty Acid Transport (GO:0015)	0.19127927	23	1.500e-03	2.733e-01	SLC27A4:37 CD36:187 FABP9:317 MFSD2A:341 ABCD3:374 PPARG:889
Proton Motive Force-Driven ATP Synthesis	-0.12921419	51	1.423e-03	2.733e-01	ATP5PD:113 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450 ATP5MG:452
Very Long-Chain Fatty Acid Catabolic Pro	0.19239333	6	1.559e-03	2.733e-01	SLC27A4:37 ABCD3:374 ACOX1:1527 ABCD2:2568 SLC27A2:3576 ABCD4:3898
Metal Ion Transport (GO:0030001)	0.07246803	157	1.775e-03	2.876e-01	KCNK18:30 CLDN16:45 ATP2A1:47 KCNJ16:53 KCNA4:114 TRPM7:169
Positive Regulation Of Cytokine Producti	-0.05519815	272	1.815e-03	2.876e-01	TRIM15:1 CCL1:38 BRCA1:51 NLRP9:64 NFAM1:71 IL6ST:117
Protein Stabilization (GO:0050821)	-0.06677090	182	1.952e-03	3.004e-01	PP1B:188 CPN2:192 PHB2:268 APOA2:294 STK3:357 MDM4:429
Sodium Ion Transmembrane Transport (GO:0	0.10125709	78	2.015e-03	3.015e-01	SLC6A6:115 SLC6A15:192 SCNN1G:263 NALCN:377 SLC6A19:402 SLC6A14:588
Regulation Of Cilium Beat Frequency (GO:	-0.39556835	5	2.189e-03	3.188e-01	CCDC40:29 CYB5D1:398 CFAP43:687 CFAP260:853 DNAH1:6452 NA
Centromere Complex Assembly (GO:0034508)	-0.24406899	13	2.314e-03	3.196e-01	MIS12:383 DLGAP5:P595 HUJRP:685 CENPC:738 MIS18A:1051 OIP5:1146
Receptor Signaling Pathway Via STAT (GO:	-0.16091162	30	2.294e-03	3.196e-01	CSF2RB:15 GHR:574 STAT2:579 JAK3:663 MYO1C:777 COR2:821
Neutral Amino Acid Transport (GO:0015804)	0.14854804	34	2.733e-03	3.681e-01	SLC1A1:64 SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A14:588

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS	0.18717674	101	8.300e-11	5.385e-07	SLC1A1:64 SLC43A1:68 SLC4A3:112 SLC6A6:115 SLC4A1:146 SLC6A15:192
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.09595441	236	4.030e-07	1.307e-03	SLC27A4:37 SLC1A1:64 SLC43A1:68 SLC4A3:112 SLC6A6:115 SLC4A1:146
BENPORATH_ES_WITH_H3K27ME3	0.04455657	978	2.961e-06	6.404e-03	RG59:20 DUSP8:43 LYSMD2:71 KCNQ5:125 PADI2:150 SNTB1:156
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.23208494	31	7.760e-06	1.259e-02	SLC43A1:68 SLC6A6:115 SLC6A15:192 SLC36A2:358 SLC6A19:402 SLC6A14:588
REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R	-0.09962129	149	2.762e-05	2.987e-02	ATP5PD:113 LRPPRC:144 LDHC:157 NDUFA11:192 NDUFB10:305 ME2:313
REACTOME_SLC_TRANSPORTER_DISORDERS	0.12893282	90	2.391e-05	2.987e-02	NUP160:14 SLC27A4:37 SLC1A1:64 SLC4A1:146 SLC4A0A1:222 NUP210:238
NIKOLSKY_BREAST_CANCER_5P15_AMPICON	0.25083793	22	2.464e-05	3.764e-02	SLC6A19:402 LPCA1:504 CEP72:904 AHRH:975 SDHA:1104 EXOC3:1164
FLORIO_HUMAN_NEOCORTEX	-0.33750187	12	5.158e-05	3.764e-02	FCRL4:87 FUT6:108 ACSM2B:524 FAM17B:577 PIWIL3:769 ERAP2:987
REACTOME_CHROMOSOME_MAINTENANCE	-0.11894923	97	5.221e-05	3.764e-02	CENPU:82 ATRX:174 POLR2H:215 MIS18BP1:472 HUJRP:685 TERF1:688
REACTOME_DEFENSINS	-0.26378101	19	6.878e-05	4.119e-02	DEFB132:150 TLR1:166 DEFB136:204 CCR2:821 DEFB129:1005 CCR6:1049
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.04572783	658	6.983e-05	4.119e-02	ATP10A:15 ATP6V0A2:16 ALB:25 SLC27A4:37 ATP2A1:47 ABCA5:54
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.02633560	330	1.040e-04	5.625e-02	ST6GAL2:66 ANKRD34B:104 FOXF2:172 GHSR:193 FFAR4:201 TBX21:248
REACTOME_BETA_DEFENSINS	-0.27498560	16	1.400e-04	6.486e-02	DEFB132:150 TLR1:166 DEFB136:204 CCR2:821 DEFB129:1005 CCR6:1049
REACTOME_NA_CL_DEPENDENT_NEUROTRANSMITTE	0.25353156	19	1.304e-04	6.486e-02	SLC6A6:115 SLC6A15:192 SLC22A1:240 SLC6A19:402 SLC6A14:588 SLC6A1:768
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	0.05271389	419	2.266e-04	8.871e-02	FAR2:3 PODXL:23 SKOR1:69 ANK1:144 MYO16:181 TBX21:248
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.10860445	101	2.105e-04	8.871e-02	ATP5PD:113 LRPPRC:144 NDUFA11:192 NDUFB10:305 NDUFV3:360 ATP5PB:450
WEBER_METHYLATED_HCP_IN_SPERM_DN	0.20656871	26	2.324e-04	8.871e-02	GT2FAL1:452 TULP2:457 STK31:458 SOX30:526 DMRT1:688 RBMXL2:710
REACTOME_DEPOSITION_OF_NEW_CENPA_CONTAIN	-0.16945241	39	2.511e-04	9.051e-02	CENPU:82 MIS18BP1:472 HUJRP:685 CENPC:738 CENPS:746 H2AC:1012
REACTOME_SPERM_MOTILITY_AND_TAXES	-0.37117847	8	2.709e-04	9.251e-02	CATSPERB:33 CATSPER3:197 CATSPER4:491 CATSPERG:963 CATSPERD:976 CATSPER2:1040
BENPORATH_PRC2_TARGETS	0.04431511	577	2.854e-04	9.257e-02	LYSMD2:71 GHSR:193 FFAR4:201 RIMBP3B:229 CDC:230 TBX21:248
REACTOME_ANTIMICROBIAL_PEPTIDES	-0.16260762	41	3.158e-04	9.314e-02	DEFB132:150 TLR1:166 DEFB136:204 RNASEH:490 CCR2:821 ELANE:922
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	-0.15213421	47	3.092e-04	9.314e-02	NDUFB10:305 NDUFV3:360 NDUFA8:471 TMEM126B:500 NDUFB8:505 TMEM70:628
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.04243617	553	6.859e-04	9.314e-01	SKOR1:69 CRACR2B:80 PADI2:150 SPEN:161 DLEU7:185 RUNDC3A:191
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN	0.39721210	6	7.525e-04	1.810e-01	SLC4A1:146 CA2:392 CAC1:1071 RHAG:1725 CA4:1815 AQP1:4732
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	-0.04032492	607	7.407e-04	1.810e-01	BIRC3:3 CSF2RB:15 TNFRSF6B:21 GH2:45 NEDD4:53 LCK:63
REACTOME_REPRODUCTION	-0.10132104	95	6.485e-04	1.810e-01	CATSPERB:33 BRCA1:51 BRCA2:137 CATSPER3:197 CATSPER4:491 MLH3:502
WOO_LIVER_CANCER_RECURRENCE_DN	0.11908925	67	7.534e-04	1.810e-01	SARDH:31 GCDH:127 MAT1A:140 MTPP:371 BPHL:527 ACAD5B:621
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.10801874	81	7.827e-04	1.814e-01	LRPPRC:144 NDUFA11:192 NDUFB10:305 NDUFV3:360 NDUFA8:471 TMEM126B:500
RODRIGUES_THYROID_CARINOMA_POORLY_DIFFE	0.03740577	694	8.472e-04	1.895e-01	CTNND1:11 HIRA:12 NT5C3B:57 AB2:59 SLC1A1:64 ST6GAL2:66
SERVITJA_LIVER_HNF1A_TARGETS_UP	0.08696931	121	9.633e-04	1.925e-01	MGST3:353 PCDH20:462 PKOX:602 BUB1:603 CYP17A1:609 CYP17A1:647
REACTOME_COMPLEX_I_BIOGENESIS	-0.13756652	48	9.792e-04	1.925e-01	NDUFA11:192 NDUFB10:305 NDUFV3:360 NDUFA8:471 TMEM126B:500 NDUFB8:505
WP_PROXIMAL_TUBULE_TRANSPORT	0.13691501	49	9.174e-04	1.925e-01	SLC1A1:64 SLC36A2:358 CA2:392 SLC6A19:402 SLC5A8:415 SLC34A1:974
YOSHIMURA_MAPK8_TARGETS_UP	0.02976382	1104	9.578e-04	1.925e-01	GSY2:13 RG59:20 CLPS:22 PNLP1RP1:41 KCNJ16:53 SLC1A1:64
REACTOME_METABOLISM_OF_LIPIDS	0.02372396	677	1.027e-03	1.960e-01	FAR2:3 ALB:25 MORC2:33 GPAT2:49 EC12:79 MOGAT2:89
ACEVEDO_METHYLATED_IN_LIVER_CANCER_DN	0.03830330	634	1.066e-03	1.976e-01	FGD5:2 SLC43A1:68 OTUD9:91 PPP1R12B:97 NCOA4:132 GRAM2C:133
REACTOME_INHIBITION_OF_DNA_RECOMBINATION	-0.16281104	32	1.437e-03	2.590e-01	ATRX:174 POLR2H:215 TERF1:688 TNF2:887 POLR2G:992 H2AC:1012
REACTOME_FERTILIZATION	-0.19737035	21	1.742e-03	3.055e-01	CATSPERB:33 CATSPER3:197 CATSPER4:491 ADAM2:1564 CATSPERG:963 CATSPERD:976
REACTOME_MICRORNA_MIRNA_BIOGENESIS	-0.20067040	20	1.892e-03	3.230e-01	POLR2H:215 AGO4:644 PNLR2G:992 BCND3D:1132 POLR2F:1291 XP05:1315
NIKOLSKY_BREAST_CANCER_7P22_AMPICON	0.14524016	38	1.951e-03	3.245e-01	BRAT1:78 SUN1:499 TMEM184A:516 FBXL18:534 PAPOLB:1283 UNCLN:1410
GRANDVAUX_IRF3_TARGETS_DN	0.22123736	16	2.185e-03	3.394e-01	TIMP3:1032 TYRPI1:1301 SORL1:1457 ANXA4:1878 GEM1:1920 RASA1:2013

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hypercalcaemia	0.17696804	44	4.923e-05	2.896e-01	CLDN16:45 TRPV6:186 KCNJ1:213 PTH:218 WNK4:354 PRPF4B:381
Nephrocalcinosis	0.15812849	54	5.904e-05	2.896e-01	CLDN16:45 SDS:134 SLC4A1:146 KCNJ1:213 PTH:218 HOGA1:284
Pediatric Obesity	0.13251960	65	2.229e-04	7.290e-01	VEGFA:334 AMY1C:364 LEP:417 ACAD8:469 MCHR2:729 GHR1L:739
Anemia, hereditary spherocytic hemolytic	0.43537186	5	7.474e-04	7.497e-01	SPTA1:38 ANK1:144 SLC4A1:146 SPTB:242 EPB42:460 NA
22q11 Deletion Syndrome	0.22869581	19	5.959e-04	7.497e-01	HIRA:12 ARVCF:277 DGCR8:503 SNAI1:1078 SEC24C:1678 COMT:1974
22q11 partial monosomy syndrome	0.30012234	11	6.678e-04	7.497e-01	HIRA:12 ARVCF:277 PAKKA:894 SEC24C:1678 COMT:1974 RREB1:2357
Asymmetric crying face association	0.26446634	13	9.621e-04	7.497e-01	HIRA:12 CASP3:205 ARVCF:277 SEC24C:1678 COMT:1974 RREB1:2357
Atherogenesis	0.14015211	44	4.305e-03	7.497e-01	ESR1:265 NPLC1:288 VEGFA:334 ALOX5:380 PTGDS:443 HRH1:616
Bardet-Biedl Syndrome	0.13545290	55	5.162e-04	7.497e-01	VEGFA:334 LEP:417 LZTF1L1:612 UCBX1:643 AZIN1:824 TMEM67:1144
Distal Renal Tubular Acidosis	0.24468953	14	1.526e-03	7.497e-01	ALB:25 SLC4A1:146 CA2:392 OSGEP:746 SLC26A7:1170 EPBA3:1681
Electroretinogram abnormal	0.08085884	130	1.487e-03	7.497e-01	RNF216:27 RD3:174 PROM1:224 CRB1:347 KCNJ13:368 KIZ:444
Female infertility	0.17207436	28	1.629e-03	7.497e-01	STAG3:81 ESR1:265 LEP:417 ESR2:1022 LEPR:1108 NR1I2:1111
Hyperactive renin-angiotensin system	0.33228547	8	1.136e-03	7.497e-01	KCNJ1:213 SCNN1G:263 SLC12A1:810 SCNN1B:911 CLCNKB:2079 NR3C2:3065
Increased plasma renin activity	0.33228547	8	1.136e-03	7.497e-01	KCNJ1:213 SCNN1G:263 SLC12A1:810 SCNN1B:911 CLCNKB:2079 NR3C2:3065
Kidney Calculi	0.13241761	53	8.610e-04	7.497e-01	CLDN16:45 SLC4A1:146 TRPM7:169 PTH:218 LRPA1:359 CLCN5:468
Mitochondrial Diseases	-0.04922224	346	1.758e-03	7.497e-01	LRPPRC:144 ASPM:187 NDUFA11:192 ATAD3B:195 C3:253 NDUFB10:305
Olivopontocerebellar Atrophies	0.21151936	19	1.416e-03	7.497e-01	SLC1A1:64 CASP3:205 ERCC6:484 GRIA2:634 GRIA3:738 GRM3:818
Polydipsia	0.17368184	28	1.476e-03	7.497e-01	CLDN16:45 KCNJ1:213 LEP:417 SLC12A3:531 CDC73:595 CASR:796
Retinal Diseases	0.05182931	313	1.703e-03	7.497e-01	CNTF:5 RG59:20 ALB:25 SLC6A6:115 RD